Demonstration of Proposed Features

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	**Predictive Crop Fit Optimizer (Scenario 1)**	Automatically processes seven distinctive input fields against an active inference pipeline to match soil values with optimal crops.	Implemented	Yes	Demonstrated by choosing the Scenario 1 tab in index.html .
2	**Target Crop Suitability Auditor (Scenario 2)**	cross-references a user's chosen crop target with real-time variables to intercept environmental compatibility anomalies.	Implemented	Yes	Verified via the Scenario 2 tab, successfully displaying precise parameter mismatch notifications when parameters fall out-of-bounds.
3	**Macro Policy & Analysis Engine (Scenario 3)**	Maps data arrays to macro environmental guidelines to build sustainability frameworks for regional planners.	Implemented	Yes	Triggered via the Scenario 3 tab, rendering dynamic resource efficiency roadmap blocks based on input chemistry vectors.
4	**Multi-Model Algorithmic Ranker**	Explicit loop structures evaluating Logistic Regression, KNN, Decision Tree, and Random Forest classifier variations.	Implemented	Yes	Executed in the system terminal terminal loop via train_model.py, printing explicit comparative accuracy scoring metrics.
5	Dynamic Contextual Results Panel	Modern, card-based interface detailing inputs alongside dynamic solution summaries.	Implemented	Yes	Renders dynamically via the Jinja2 template gateway engine within result.html instantly upon form execution.
6	Agro-Ecological Range Sanitizer	Strict boundary parsing loop running on the server backend to catch out-of-range numerical anomalies.	Implemented	Yes	Evaluated via boundary testing; correctly flags bad inputs with clear HTTP 400 validation messages rather than crashing.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%